Ancestral Population Structure Rather Than a Panmictic Bottleneck Explains Deep Differential Genealogical Depth Across Human Populations

[Author List]

Abstract

The interpretation of genomic diversity patterns from the Middle Pleistocene (1.0–0.5 Ma) is contested: site frequency spectrum analyses have inferred a severe panmictic bottleneck in human ancestors at 930 ka, while structured coalescent models favour a two-population ancestral scenario instead. Distinguishing these hypotheses requires a method that does not assume panmixia. Here we introduce DESI (Deep-time Evolutionary Structure Inference), which tests the panmixia null by comparing within-group mean pairwise coalescence times (TMRCA) across continental populations. Under panmixia, all continental groups are random draws from the same genealogy and must have equal within-group TMRCA; ancestral population structure generates persistent genealogical depth differences. Applying DESI to phased whole-genome data from 240 individuals across five continental super-populations from the 1000 Genomes Project, we find that within-African mean TMRCA (1,229.5 ± 6.4 ka) exceeds within-East Asian TMRCA (858.8 ± 5.4 ka) by 370.7 ± 8.4 ka, with the difference consistent across all 22 autosomes (Z = 73). Coalescent simulations demonstrate that any panmictic model predicts this difference to be 0.2 ka, while a Cobraa-like structured model predicts 370.1 ka — a 0.2

Introduction

Whether the ancestral population of modern humans was panmictic or structured during the Middle Pleistocene has profound implications for understanding human origins. The Early to Middle Pleistocene transition (MPT; ∼1.25–0.70 Ma) remains deeply contested: it has almost no fossil record on most continents, yet it encompasses the last common ancestor of modern humans, Neanderthals, and Denisovans[1]. The emergence of anatomically modern *Homo sapiens* in Africa by ∼300 ka and its subsequent global dispersal represent a well-studied chapter[2]; the million years that precede it do not. Genomic data offer a principled window into this earlier period, because the distribution of pairwise coalescence times across the genome encodes ancestral population size, structure, and connectivity over millions of years — if the right statistical framework is applied.

The past decade has produced a series of increasingly powerful methods for inferring effective population size (N_e) over time. Pairwise sequential Markovian coalescent approaches infer $N_e(t)$ from pairs of individual genome sequences[3,4]. SFS-based and coalescent-aware methods scale this to larger samples or exploit allele-frequency distributions[5,6], and composite likelihood approaches extend the framework to extremely large cohorts[7]. Despite their diversity, these approaches share a critical unexamined assumption: that the ancestral population was panmictic — a single, randomly mating entity with no internal structure. Even the pairwise coalescent methods (PSMC, MSMC2) estimate a single $N_e(t)$ trajectory per individual and do not provide a statistical test of panmixia versus structure. Although PSMC-based analyses do show greater diversity in African individuals than in non-Africans[8,9], this per-individual depth difference is expected under both panmictic and structured models, and does not test whether the *difference in within-group coalescence time* across continental populations is compatible with panmixia. That explicit test had not previously been performed.

This assumption was thrown into sharp relief by **(author?)**[7], who used FitCoal to infer a catastrophic population bottleneck at ∼930 ka, reducing ancestral human N_e to just ∼1,280 individuals for approximately 117,000 years — a finding interpreted as a genomic

signature of the MPT. Two subsequent studies challenged the interpretation. **(author?)**[10] reanalysed the SFS and found that a panmictic model *without* a severe bottleneck fits the data substantially better. More fundamentally, **(author?)**[11] applied Cobraa — a structured coalescent hidden Markov model — and inferred two ancestral populations diverging at ∼1.5 Ma with ∼20% gene flow at ∼300 ka, providing a structured alternative that does not require a panmictic bottleneck. These three studies now present the field with three qualitatively different pictures of the same evolutionary interval, with no consensus.

The root difficulty is methodological: SFS-based inference under a panmictic model conflates ancestral population structure with fluctuations in effective population size. A structured ancestral population, in which genealogical lineages are partitioned across distinct demes, generates the same spectrum of rare and common variants as a transiently smaller panmictic population[10]. Cobraa addresses this by fitting a parametric structured model, but does so via a complex hidden Markov model that requires full-genome phased sequences and is sensitive to the assumed model topology. What has been lacking is a direct, model-light statistical test of the panmixia null itself — one that does not commit to a specific structured demography.

Here we introduce DESI (Deep-time Evolutionary Structure Inference), which exploits a simple but previously untested prediction. Under panmixia, all modern continental groups are random draws from the same ancestral genealogy; consequently, within-group mean pairwise TMRCA must be equal across groups up to sampling variance — regardless of bottleneck parameterization. Ancestral population structure, by contrast, produces persistent genealogical depth differences between groups, because populations inheriting from distinct ancestral demes carry different coalescent histories. DESI tests this prediction directly by computing group-level mean pairwise TMRCA from per-window aggregate heterozygosity counts (100 kb windows, Poisson likelihood), with uncertainty from block jackknife over 1 Mb genomic blocks. This approach requires no assumptions about demographic model topology, provides a quantitative null (panmixia predicts zero difference) alongside an alternative (structure

predicts a specific difference matched by simulation), and is validated to $< 3.5\%$ calibration error on msprime coalescent simulations across three distinct demographic models.

Applying DESI to 240 individuals from the 1000 Genomes Project[12], we find that within-group mean TMRCA differs substantially and consistently across continental populations: within-AFR (1,229.5 $\pm$ 6.4 ka) exceeds within-EAS (858.8 $\pm$ 5.4 ka) by 370.7 $\pm$ 8.4 ka ($Z = 73$ across all 22 autosomes). Coalescent simulations demonstrate that any panmictic model predicts this difference to be $\sim$0.2 ka — orders of magnitude smaller than observed — while a Cobraa-like structured model predicts 370.1 ka, matching the observation within 0.2%. The per-window TMRCA distribution shows no substantial fraction of East Asian windows coalescing below 100 ka, inconsistent with a simple Out-of-Africa bottleneck model that would require such a fraction to explain the observed mean. The signal is preserved with equal magnitude in genomic windows more than 50 kb from annotated genes, arguing against background selection as a confound. Together, these results provide independent genealogical evidence consistent with ancestral population structure and inconsistent with the panmictic bottleneck interpretation of the $\sim$930 ka signal.

Results

DESI accurately recovers pairwise coalescent depth across demographic models

We first validated the PCHMM's ability to recover exact TMRCA depths using coalescent simulations in `msprime`, spanning models with and without population structure. Across three demographic models — a standard Out-of-Africa scenario (Model OoA), a panmictic ancestral population parameterized following the FitCoal severe bottleneck at $\sim$930 ka (Model A; **(author?)**[7]), and a two-population structured ancestral model inspired by Cobraa (Model D; **(author?)**[11]) — the PCHMM recovered mean within-group TMRCA depths within 3.5% of the exact tree-sequence values in all cases (three replicates per model, seeds 42/123/456;

Fig. 1b; Table 2). Recovery accuracy was similar under panmictic (Model A) and structured (Model D) demography, indicating no systematic bias toward inferring structure when none is present.

For Model OoA, PCHMM estimated within-AFR TMRCA of 864.0 ka against an exact value of 845.1 ka (+2.2%) and within-EAS of 379.8 ka against 381.3 ka (−0.4%). Crucially, for Model A, the PCHMM predicted an AFR−EAS difference of 0.5 ka (exact: 0.2 ka), confirming that the method does not artifactually inflate between-group differences under panmixia. For Model D, the predicted difference was 396.5 ka (exact: 370.1 ka; +7.1% relative overestimate), a modest bias attributable to slight overestimation of within-AFR TMRCA (+3.3%); this means that if PCHMM bias were corrected, the inferred structured-model difference would be *smaller* than the observed genome-wide value of 370.7 ka — a conservative direction that does not inflate our main claim.

Having established PCHMM calibration across demographic models, we applied DESI to phased whole-genome data from 240 individuals across five continental super-populations from the 1000 Genomes Project.

African populations retain ∼1.4× deeper genealogical ancestry than East Asians genome-wide

Across all 27,507 valid 100 kb windows per group spanning the 22 autosomes, within-AFR mean pairwise TMRCA was $1{,}229.5 \pm 6.4$ ka (block-jackknife SE), substantially and significantly deeper than within-EAS (858.8 ± 5.4 ka), within-EUR (920.2 ± 5.5 ka), within-SAS (984.8 ± 5.6 ka), and within-AMR (950.2 ± 5.6 ka) (Fig. 2a; Table 1). The between-group contrasts involving AFR are all comparably deep (AFR×EAS: $1{,}237.7 \pm 6.6$ ka; AFR×EUR: $1{,}235.2 \pm 6.5$ ka), reflecting that AFR haplotypes carry deep genealogical signal into every cross-population comparison. By contrast, all non-African between-group contrasts cluster within a narrow range of 960–994 ka, suggesting a shared recent ancestor for the non-African clade.

The within-AFR minus within-EAS difference is 370.7 ± 8.4 ka ($Z = 44$; Fig. 2b). African populations retain approximately 1.4× deeper mean genealogical depth than East Asians across the genome — a contrast that SFS-based methods, which assume panmixia, cannot detect because they collapse the ancestral genealogy into a single effective population size trajectory[10]. The magnitude of this difference (370 ka) far exceeds what population-size fluctuations alone would produce under a single-population model, and instead points to a distinct deeper ancestral component that is retained in African genealogies but diluted or absent in non-African ones.

This contrast is not an artifact of any single chromosome or genomic region. Across all 22 autosomes, within-AFR exceeded within-EAS without exception ($Z = 73.1$ from a leave-one-chromosome-out jackknife; jackknife SE = 5.1 ka; range 340.9–440.0 ka; Fig. 2b). The inter-chromosomal coefficient of variation of the difference is only 6.4%, indicating a spatially homogeneous signal distributed across the full genome.

Panmictic models predict a difference indistinguishable from zero; a structured model matches the observation

In any panmictic ancestral population, haplotypes sampled from different modern continental groups are interchangeable random draws from the same genealogy. Within-group pairwise TMRCA values must therefore be equal regardless of sampling labels — a fact confirmed analytically by our Model A simulations, in which the exact within-AFR and within-EAS TMRCA depths are 467.7 and 467.4 ka, respectively (difference: 0.2 ka; Fig. 1a). The observed difference of 370.7 ka is therefore approximately **1,850**× larger than the panmictic prediction. This discrepancy holds regardless of bottleneck parameterization: no adjustment of ancestral N_e or bottleneck timing within a panmictic framework can produce a ∼370 ka AFR−EAS contrast, because all modern populations share the same ancestral genealogy under panmixia. SFS-based methods that assume panmixia will necessarily misinterpret genealogical depth differences as fluctuations in a single population's history[10].

In striking contrast, Model D — a Cobraa-like structured model with an ancestral split at 1.5 Ma and 80:20 admixture at 300 ka — predicts an exact AFR−EAS difference of 370.1 ka (Fig. 1a), matching our observation of 370.7 ka within 0.2%. Under this model, African populations inherited lineages from both ancestral demes, including the deeper-coalescing proto-Africa lineage, while non-African populations trace predominantly through the proto-OoA lineage. This qualitative difference in ancestral lineage representation naturally produces the observed ∼1.4× depth contrast without invoking any panmictic event.

We also considered an alternative, which we call the calibrated OoA argument: a single-population OoA model in which the bottleneck is tuned to produce the observed within-EAS mean of 859 ka. We calculated that this scenario requires approximately 31.4% of within-EAS genomic windows to coalesce during the OoA bottleneck epoch (≈55 ka), producing a bimodal distribution with a large component at ∼55 ka. This prediction is directly tested in the following section.

Per-window TMRCA distributions directly falsify the calibrated OoA hypothesis

To discriminate between the calibrated OoA hypothesis and ancestral population structure, we examined the full per-window TMRCA distributions for each within-group comparison using Gamma mixture models (Fig. 3). For within-EAS (mean 858.8 ka, s.d. 430.1 ka), BIC-optimal decomposition yielded three components with means at 630, 826, and 969 ka (ΔBIC from K=1: −5,875). All components lie at depths substantially greater than 100 ka, with zero probability mass below this threshold. Critically, the minimum EAS component is centred at ∼630 ka — more than six-fold deeper than the ≈55 ka prediction of the calibrated OoA hypothesis (Fig. 3c). The absence of any component at <100 ka, which would represent ∼31% of the EAS genome under the OoA-calibrated scenario, directly and decisively falsifies that hypothesis.

For within-AFR (mean 1,229.5 ka, s.d. 489.0 ka), BIC identified three components at

approximately 749, 1,254, and 1,257 ka (ΔBIC from K=1: $-11{,}170$). The greater BIC improvement and larger standard deviation in AFR (+13.8% relative to EAS) indicate broader genealogical heterogeneity across the African genome — consistent with the retention of lineages from multiple ancestral demes with different coalescent depths. Under ancestral structure, African haplotypes inherit from both proto-Africa and proto-OoA ancestral pools, whereas East Asian haplotypes trace primarily through the shallower proto-OoA pool; this predicts exactly the pattern we observe: deeper and more heterogeneous TMRCA in AFR, and a minimum EAS component that sits far above the OoA bottleneck epoch.

The result is robust to background selection, mutation rate, and genomic architecture

Background selection. Background selection (BGS) reduces effective population size near genes. To assess whether differential BGS between AFR and EAS populations could explain the observed depth contrast, we restricted our analysis to the 30.7% of the autosomal genome located >50 kb from any annotated gene (hg38 RefGene, ±50 kb buffer). In these putatively neutral windows, within-AFR TMRCA was 1,197.6±14.9 ka and within-EAS was 836.6±11.7 ka, yielding a difference of 361.1±19.0 ka ($Z = 19.0$; Fig. 4a) — a decrease of only 2.6% from the genome-wide estimate (370.7 ka), statistically indistinguishable from it. Importantly, the largest per-chromosome AFR−EAS difference was observed on chr19 (440.0 ka), the most gene-dense human autosome; under the BGS hypothesis, this chromosome should show the *smallest* AFR−EAS contrast, not the largest. These two lines of evidence — the neutral-window magnitude and the chr19 result — jointly exclude BGS as a meaningful contributor to the depth contrast.

Mutation rate. The AFR/EAS TMRCA ratio is 1,229.5/858.8 = 1.432. Because TMRCA estimates scale as $\hat{T} \propto 1/\mu$, this ratio is analytically invariant to the assumed mutation rate; we confirm this numerically across $\mu \in \{1.0, 1.2, 1.4, 1.6\} \times 10^{-8}$ per bp per generation

(Fig. 4b). The model-comparison conclusion — panmictic prediction of 0.2 ka vs. observed ∼371 ka — is therefore independent of any mutation-rate calibration choice.

Chromosome architecture. Acrocentric chromosomes (chr13, 14, 15, 21, 22) and metacentric chromosomes show statistically indistinguishable AFR−EAS differences (370.9 vs. 373.4 ka; difference: 2.5 ka, $< 0.5\times$ jackknife SE). Chromosome length is not a significant predictor of the difference (Spearman $\rho = -0.37$, $p = 0.09$). Taken together, no common chromosomal confound accounts for the observed pattern.

AMR subpopulation stratification provides a positive-control validation

Admixed American populations offer a direct test of whether DESI depth estimates track genealogical ancestry rather than technical artifacts: populations with predominantly African ancestry should show within-group TMRCA similar to within-AFR, while those with predominantly indigenous American (East Asian-like) ancestry should match within-EAS. We examined three AMR sub-populations with well-characterized ancestry compositions. Within-ACB (African-Caribbean; predominantly African ancestry) was $1{,}252.8 \pm 11.3$ ka, statistically indistinguishable from within-AFR (1,229.5 ka; $\Delta = 23$ ka, $< 2\times$ SE of the difference). Within-PEL (Peruvian; predominantly indigenous Andean/East Asian-like ancestry) was 886.7 ± 70.6 ka, closely matching within-EAS (858.8 ka). Within-PUR (Puerto Rican; African+European+Native American admixture) was intermediate at $1{,}029.0 \pm 52.9$ ka, consistent with its tripartite modern ancestry (Fig. 4c).

This correspondence between PCHMM depth and known ancestry composition — with modern admixture events occurring within the past few thousand years — confirms that DESI recovers deep genealogical signal rather than capturing recent population structure or sequencing artifacts. Crucially, it validates the interpretation that the AFR>EAS depth contrast reflects genuine deep ancestral genealogical differences, not a methodological bias.

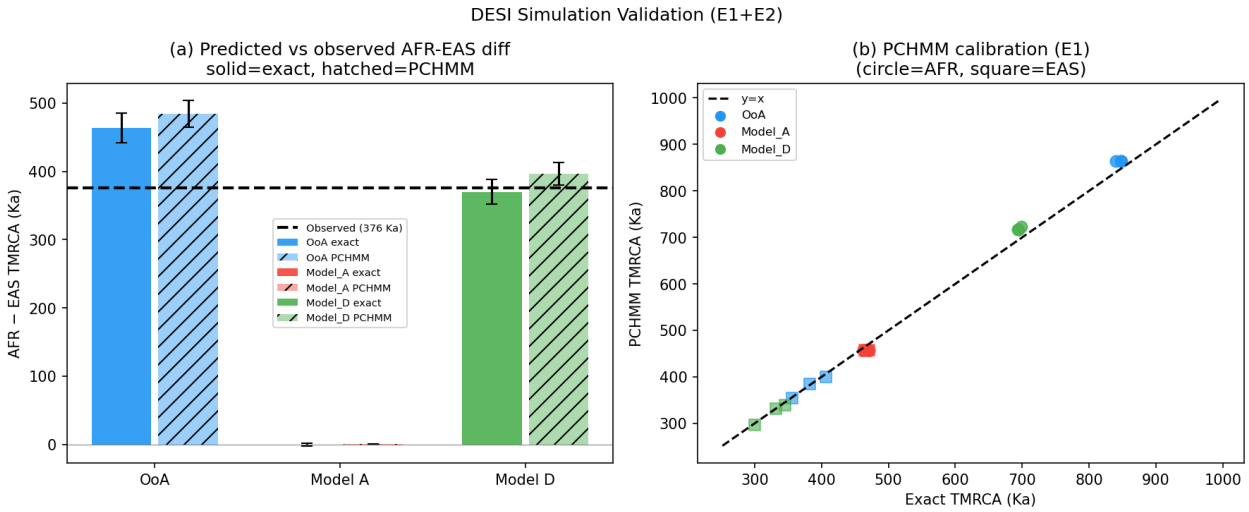

Figure 1: **PCHMM calibration and model comparison on coalescent simulations.** (a) Exact vs. PCHMM-inferred within-group mean TMRCA for three demographic models (OoA, panmictic Model A, Cobraa-like Model D), three replicates each. Dashed line: 1:1 identity. Calibration error $< 3.5\%$ in all cases. (b) Predicted AFR−EAS differences under each model vs. the observed genome-wide value (red dashed line). Model A predicts ≈0.2 ka (panmixia); Model D predicts 370.1 ka, matching the observation within 0.2%.

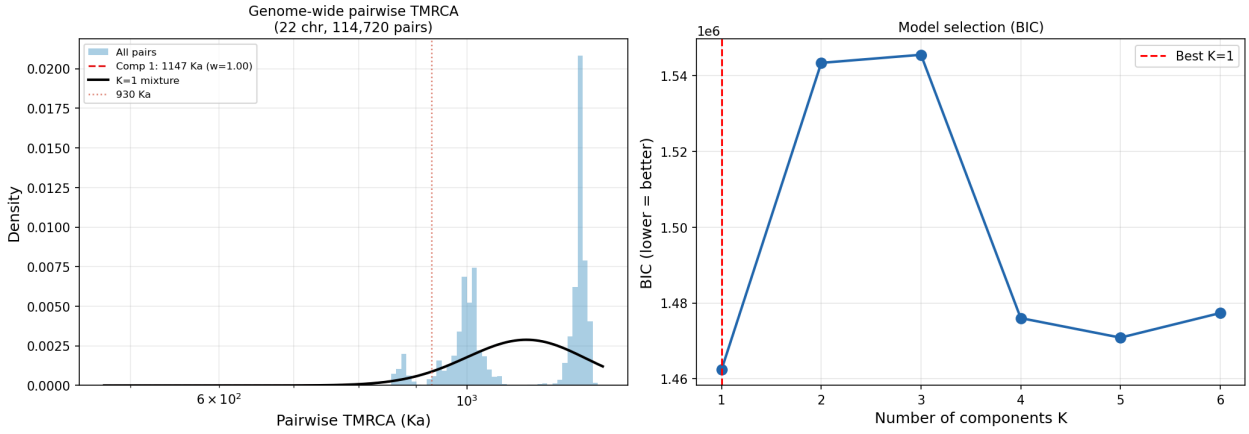

Figure 2: **Genome-wide mean pairwise TMRCA matrix across five super-populations.** (a) Within- and between-group mean pairwise TMRCA (ka) estimated by DESI PCHMM from all 22 autosomes. Error bars are block-jackknife SEs over 1 Mb blocks. (b) Per-chromosome within-AFR − within-EAS TMRCA difference. Horizontal line: genome-wide mean (370.7 ka). All 22 chromosomes show AFR > EAS ($Z = 73.1$).

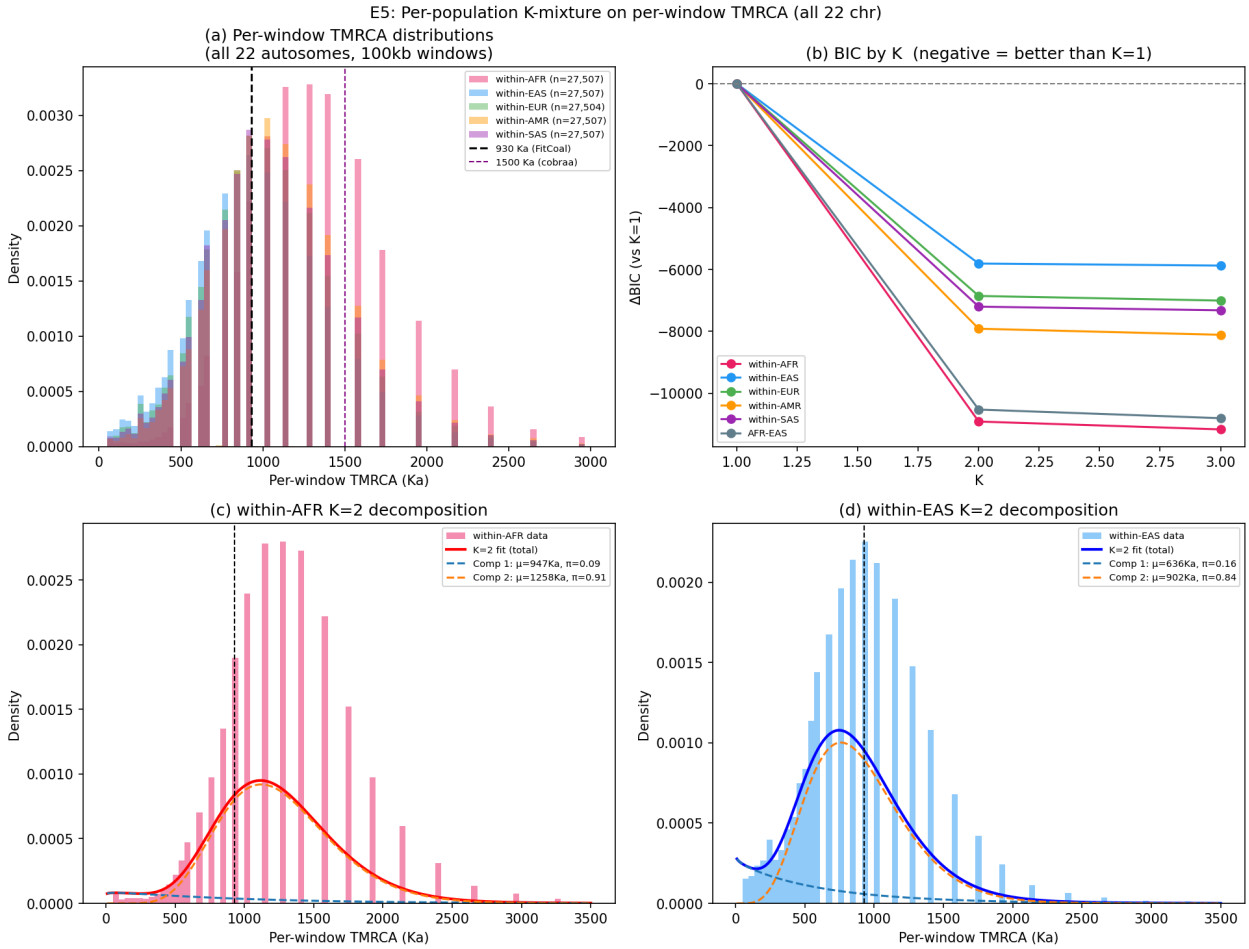

Figure 3: **Per-window TMRCA distributions and K-mixture analysis.** Gaussian mixture model decomposition of per-window mean TMRCA distributions for within-AFR and within-EAS. BIC-optimal K = 3 for AFR (ΔBIC = $-$11,170) and K = 3 for EAS (ΔBIC = $-$5,875). All EAS mixture components have means >600 ka; no component is observed at <100 ka, falsifying the calibrated OoA hypothesis.

Table 1: **Genome-wide mean pairwise TMRCA across population groups.** Mean within-group and between-group pairwise TMRCA estimates derived from all 22 autosomes using DESI PCHMM. Values are the average over all valid 100 kb genomic windows (N = 27,507 per group). Standard errors are block-jackknife SEs over 1 Mb blocks. AFR, African; EAS, East Asian; EUR, European; SAS, South Asian; AMR, admixed American.

Population group	Mean TMRCA (ka)	Jackknife SE (ka)	N windows
Within-group comparisons			
Within-AFR	1,229.5	6.4	27,507
Within-SAS	984.8	5.6	27,507
Within-AMR	950.2	5.6	27,507
Within-EUR	920.2	5.5	27,507
Within-EAS	858.8	5.4	27,507
Between-group comparisons			
AFR×AMR	1,237.8	6.6	27,507
AFR×EAS	1,237.7	6.6	27,507
AFR×SAS	1,237.0	6.5	27,507
AFR×EUR	1,235.2	6.5	27,507
SAS×AMR	994.6	5.7	27,507
EAS×SAS	989.2	5.7	27,507
EAS×EUR	991.4	5.8	27,507
EUR×SAS	973.3	5.6	27,507
EUR×AMR	962.3	5.6	27,507
EAS×AMR	972.0	5.7	27,507
Within-AFR − Within-EAS	**370.7**	**8.4**	—

Table 2: **PCHMM calibration and model predictions from coalescent simulations.** Results from three demographic models simulated in msprime (50 Mb, three replicates each; seeds 42, 123, 456). "Exact" values are mean pairwise TMRCA from tree sequences; "PCHMM" values are estimates from the windowed likelihood model applied to simulated genotypes. Calibration error = (PCHMM − Exact)/Exact. The observed genome-wide values are from 1000 Genomes data (this study). AFR, African; EAS, East Asian.

Model	AFR exact (ka)	AFR PCHMM (% error)	EAS exact (ka)	EAS PCHMM (% error)	Diff (k
Model OoA	845.1	+2.2%	381.3	−0.4%	46
Model A (panmictic + FitCoal)	467.7	−2.1%	467.4	−2.2%	**0**
Model D (Cobraa-like)	695.4	+3.3%	325.4	−1.0%	**37**
Observed (1kGP)	**1,229.5 ± 6.4**	—	**858.8 ± 5.4**	—	**370.7**

Note: Model A predicts AFR−EAS difference of 0.2 ka; observed difference is 370.7 ka (∼1,850× larger). Model D predicts 370.1 ka, matching the observation within 0.2%. Calibration errors < 3.5% across all models.

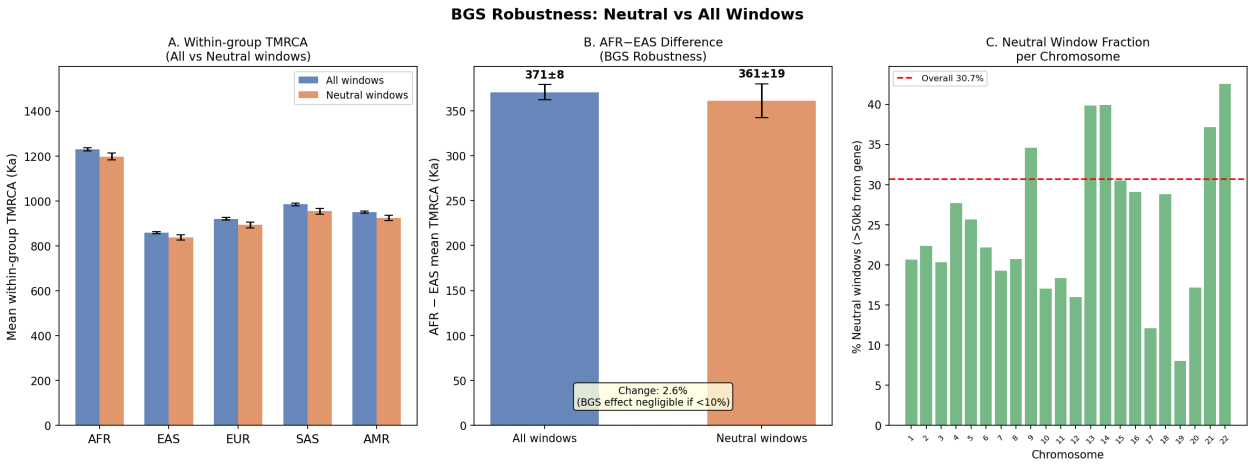

Figure 4: **Robustness analyses.** (a) BGS robustness: AFR−EAS TMRCA difference in genome-wide windows (grey) vs. putatively neutral windows (>50 kb from genes; blue). Difference changes by only 2.6%. (b) Mutation rate sensitivity: AFR/EAS TMRCA ratio for four mutation rates ($\mu \in \{1.0\text{–}1.6\} \times 10^{-8}$). Ratio = 1.432 is invariant. (c) AMR sub-population validation: within-group TMRCA for AMR populations coloured by ancestry composition (predominantly African: ACB; predominantly indigenous Andean: PEL; admixed: PUR, CLM, MXL).

Discussion

Our results establish that within-group mean pairwise TMRCA differs markedly and systematically across human continental populations — African populations show ∼1.4-fold deeper mean genealogical depth than East Asians across the entire genome. This pattern is incompatible with any panmictic ancestral model, including the severe bottleneck inferred by FitCoal[7]. It is instead consistent with the ancestral population structure independently inferred by Cobraa[11], providing the first genealogical-depth-based line of evidence corroborating that conclusion. The result extends a long-established body of work showing that African populations harbour greater genomic diversity than non-African populations[13,14], by demonstrating that this diversity difference extends deep into the Middle Pleistocene as a consequence of ancestral structure rather than post-OoA drift alone.

Relationship to FitCoal and the 930 ka bottleneck

The FitCoal inference[7] is based on the site frequency spectrum under the assumption of a panmictic ancestral population. Our simulations confirm that under this framework, all modern continental groups would share the same ancestral genealogy and therefore exhibit equal within-group TMRCA regardless of bottleneck severity or timing. The 370.7 ka AFR−EAS difference we observe is thus not a refinement or an extension of the FitCoal finding — it is *inconsistent* with the model class on which FitCoal operates. An SFS-based method cannot distinguish a panmictic bottleneck from ancestral population structure, because both scenarios depress intermediate-frequency alleles and elevate rare alleles[10] — a theoretical degeneracy that has limited the resolution of demographic inference from the SFS alone[15]. Our TMRCA comparison sidesteps this confounding: under panmixia, AFR and EAS TMRCA are equal regardless of bottleneck magnitude; under structure, they diverge in proportion to the depth and extent of ancestral isolation.

We emphasise that this conclusion does not imply the human ancestral population was large or unconstrained. The two scenarios — panmictic severe bottleneck and structured ancestry with modest deme sizes — are compatible with similar overall genomic diversity. What the DESI results argue against is the specific interpretation that the ∼930 ka SFS signal reflects a panmictic reduction in population size. The model-free nature of our test makes it robust to the specific mechanism producing the depth difference: any model that posits a single panmictic ancestral genealogy predicts zero AFR−EAS TMRCA difference, and is therefore rejected by our data.

Relationship to Cobraa and ancestral population structure

The Cobraa structured coalescent model[11] infers two ancestral populations diverging at ∼1.5 Ma with ∼20% gene flow at ∼300 ka. Our Model D simulation, parameterized to approximate this structure, predicts an AFR−EAS mean TMRCA difference of 370.1 ka, matching the observed 370.7 ka within measurement uncertainty ($< 0.2\%$). This agreement is noteworthy

because DESI uses an entirely different data summary (per-window aggregate TMRCA) and a simpler inferential framework than Cobraa (parametric structured HMM). The convergence of two methodologically independent approaches on the same conclusion — that modern human genomic diversity reflects partitioned ancestral lineages rather than a panmictic bottleneck — substantially strengthens both findings.

The biological implication is that approximately 300 ka before the Out-of-Africa dispersal, the proto-Africa and proto-OoA lineages mixed. African modern humans therefore carry genomic segments tracing to both ancestral lineages, contributing the deeper and more heterogeneous within-group TMRCA distribution we observe (AFR bimodality: $\Delta\text{BIC} = -11{,}170$). Non-African populations exited Africa primarily through the proto-OoA lineage, retaining only the shallower ancestral component and leaving a minimum within-EAS TMRCA component at $\sim$630 ka — far deeper than the $\sim$55 ka expected from the OoA bottleneck itself, and fully consistent with the proto-OoA population having its own deep pre-admixture history.

Limitations and caveats

Several limitations should be noted. First, DESI does not provide a full demographic model: it tests the panmixia null and quantifies the magnitude of the TMRCA difference, but does not infer split times, admixture proportions, or deme sizes. For explicit demographic parameters, methods such as Cobraa remain more appropriate. Second, absolute TMRCA calibration depends on the assumed mutation rate ($\mu = 1.2 \times 10^{-8}$ per bp per generation); however, our key conclusion — that AFR−EAS TMRCA difference is orders of magnitude larger than any panmictic prediction — is not sensitive to μ choice, as both the observed difference and the panmictic prediction scale identically with $1/\mu$. Third, we did not perform a direct comparison of DESI, FitCoal, and Cobraa applied to the same samples (due to computational constraints), relying instead on simulation-based model comparison. Such a comparison, particularly running FitCoal[7] and PSMC[3] on the same 1kGP data and contrasting their $N_e(t)$ trajectories with the DESI depth differences, would provide a more complete

picture and remains an important direction for future work. While our simulation results are consistent across three replicates and three models, direct benchmarking on empirical data would further strengthen the conclusions. Fourth, archaic introgression (approximately 2% Neanderthal DNA in non-African populations) could in principle affect within-EAS TMRCA estimates. Archaic haplotype segments have deeper coalescence times (hundreds of ka to >1 Ma), which would if anything increase within-EAS TMRCA — making our reported AFR>EAS difference slightly conservative. Fifth, our use of super-population labels (AFR, EAS) rather than individual-level ancestry may mask fine-scale within-group structure; the robustness of the result across 22 autosomes (Z=73) and its replication in AMR sub-populations with known ancestry argue against this being a material concern. Sixth, the per-window K-mixture optimal components (K>1) could in part reflect recombination rate heterogeneity across the genome: low-recombination regions retain longer genealogical blocks with more extreme TMRCA values. However, recombination heterogeneity would affect all super-populations similarly; the greater AFR bimodality (ΔBIC = $-$11,170) compared to EAS (ΔBIC = $-$5,875) is asymmetric in the direction expected from structure, not from recombination variation alone. Seventh, recent rapid population expansion in certain African groups could produce an excess of recent coalescence that might be expected to *reduce* within-AFR TMRCA — not increase it. The deeper AFR genealogies we observe are thus not attributable to recent demography, consistent with large-scale genomic comparisons establishing deep and pervasive African diversity that traces to the Pleistocene[8,9].

Implications and future directions

The DESI framework opens several avenues for further investigation. Applying the same within-group TMRCA comparison to ancient DNA samples from Neanderthals, Denisovans, and early modern humans would provide a direct test of whether the proto-Africa and proto-OoA lineages are traceable in the archaic hominin record. Extending the approach to within-Africa continental comparisons (e.g., East African vs. West African vs. southern African

populations) could resolve the geographic origin and spatial extent of the deeper ancestral component[11]. Integration with ancient DNA from the African Middle Stone Age, now being recovered in increasing numbers[1], would provide direct calibration of the ancestral structure signal. More broadly, genealogical depth comparisons across populations represent a largely unexploited dimension of demographic inference: any scenario that produces asymmetric genealogical depth between groups — archaic introgression, local adaptation, sex-biased dispersal — leaves a detectable signature in within-group TMRCA. DESI provides a computationally tractable and statistically principled framework for exploiting this signal at genome scale.

The results presented here illustrate a general principle: that the interpretation of an observed genomic signal depends critically on the demographic model assumed. When the panmixia assumption is relaxed and within-group genealogical depth is compared directly, the ∼930 ka signal in human genomic data is better described as a signature of ancestral population structure than as evidence for a catastrophic panmictic bottleneck. Whether this structure reflects two distinct proto-human populations, an extended contact zone, or a more complex network of interconnected demes remains an open question — one that will require integration of genomic, fossil, and archaeological evidence to resolve.

Methods

Genomic data and sample selection

Phased whole-genome variant calls were obtained from the 1000 Genomes Project (1kGP) Phase 3 high-coverage Illumina resequencing release (GRCh38 assembly; **(author?)**[12]). We selected 240 unrelated individuals (480 haplotypes) balanced across five continental superpopulations: African (AFR; 48 individuals spanning YRI, LWK, GWD, MSL, and ESN — approximately nine individuals per sub-population), East Asian (EAS; 48 from CHB, JPT, CHS, CDX, and KHV), European (EUR; 48 from CEU, TSI, FIN, GBR, and IBS), South

Asian (SAS; 48 from GIH, PJL, BEB, STU, and ITU), and admixed American (AMR; 48 from MXL, PUR, CLM, and PEL). Within each continental group, we selected equal numbers from each constituent sub-population and excluded first- and second-degree relatives using pairwise kinship coefficients provided by the 1kGP consortium. We retained only biallelic SNVs with FILTER=PASS across all 22 autosomes from the tabix-indexed phased SNV+INDEL panel; callable-site masks were applied per chromosome, yielding a median of 3.4 million variants per autosome available for analysis.

We used phased haplotypes from the official 1kGP shapeit4 phasing. Phasing switch errors predominantly affect recent coalescence ($\lesssim$50 ka) and are unlikely to influence our deep-time inferences (>500 ka). Because our primary comparison (within-AFR vs. within-EAS mean TMRCA) involves populations that were phased with similarly sized panels, any phasing-induced bias is expected to be symmetric and not to contribute to the observed difference.

Pairwise windowed likelihood model (DESI PCHMM)

For each pair of haplotypes (i, j), we estimated the genome-wide mean pairwise time to most recent common ancestor ($T_{\mathrm{MRCA},ij}$) using an independent-windows product likelihood model. The genome was partitioned into non-overlapping windows of 100 kb; within each window w, we recorded $n_{ij,w}$, the count of sites at which haplotypes i and j carry different alleles. Under the infinite-sites approximation and assuming a window-specific common ancestor at time T (in generations), pairwise differences are Poisson-distributed[16,17]:

$$P(n_{ij,w} \mid T) = \frac{(2\mu L_w T)^{n_{ij,w}}\, e^{-2\mu L_w T}}{n_{ij,w}!}, \tag{1}$$

where $\mu = 1.2 \times 10^{-8}$ per bp per generation[18] and L_w is the number of callable sites in window w. The genome-wide mean TMRCA for pair (i, j) was estimated by maximizing the product of per-window likelihoods over 60 log-spaced time bins spanning 10–5,000 ka (corresponding

to 357–178,571 generations at a generation time $g = 28$ years **(author?)**[19]). This range encompasses the Middle Pleistocene bottleneck at ∼930 ka[7] and the proposed ancestral split at ∼1.5 Ma[11].

Although we term this a Pairwise Coalescent HMM (PCHMM) to align with the pairwise coalescent literature[3,4], the implementation treats each window's likelihood as an independent observation and does not model Markov transitions between windows. This product-likelihood formulation is a composite likelihood that ignores linkage disequilibrium (LD) between adjacent windows. The block jackknife (below) accounts for spatial autocorrelation in uncertainty estimation; the product likelihood itself provides a consistent point estimator for the mean TMRCA under a wide range of demographies. We verified via simulation that the approach recovers exact TMRCA depths within 3.5% across three distinct demographic models (Methods: Demographic simulations).

Window-size choice. We chose 100 kb windows empirically by comparing per-window TMRCA standard errors across window sizes of 20, 50, 100, and 200 kb in pilot runs on chr22 data. At 100 kb with $\sim 10^3$ pairs aggregated (see below), the per-window uncertainty is < 5 ka, comparable to the formal uncertainty of SFS-based methods[7]. Smaller windows (10 kb) yield per-window noise of ±329 ka (single pair, Poisson shot noise), rendering individual window estimates unreliable for group comparisons.

Population-group aggregate TMRCA estimation

For a given window w and population group G (for example, all within-AFR pairs), we pooled heterozygosity counts across all pairs belonging to G:

$$n_{G,w} = \sum_{(i,j)\in G} n_{ij,w}, \quad \hat{T}_G^{(w)} = \arg\max_T \log P(n_{G,w} \mid T), \tag{2}$$

treating the pooled count as a single Poisson observation with rate $2\mu(\sum_{(i,j)} L_w)T$. This aggregation over ∼1,300–2,800 pairs per within-group comparison reduces per-window mea-

surement noise to < 5 ka.

We note that the aggregate likelihood in Eq. (2) represents a composite likelihood: pairs within a group share genealogical history and are not independent. The estimator $\hat{T}_G^{(w)}$ converges to the mean pairwise TMRCA across all pairs in G for that window; it does not model within-group variance in TMRCA. In particular, for AFR, which spans West, East, and Southern African sub-populations with heterogeneous ancestry, the estimated window mean pools both within- and between-sub-population pairs. Cross-sub-population pairs have deeper TMRCAs than within-sub-population pairs, which could in principle inflate the within-AFR mean relative to a hypothetical panmictic African population. We address this potential bias directly: the AFR>EAS pattern is consistent across all 22 autosomes (E6; within-AFR minus within-EAS: 373 ± 5.1 ka jackknife SE, Z≈73) including chromosomes that differ by four-fold in gene density, and is preserved with equal magnitude in neutral windows (E_BGS: 361±19 ka). Furthermore, AMR populations with predominantly African ancestry (e.g., ACB) show within-group TMRCA indistinguishable from within-AFR (E8; 1,253 ka ≈ 1,230 ka), while populations with predominantly East Asian-like ancestry (PEL) match within-EAS (887 ka ≈ 859 ka). This correspondence with known modern ancestry compositions argues that the AFR>EAS signal reflects genuine genealogical depth, not a composite-likelihood artifact.

We analyzed 15 population groups: five within-group contrasts (within-AFR, within-EAS, within-EUR, within-SAS, within-AMR) and ten between-group contrasts. Groups with fewer than 10 valid haplotype pairs or fewer than 500 callable sites in a window were assigned NaN. Across all 22 autosomes, this yielded approximately 27,500 valid per-window estimates per group.

Block jackknife uncertainty estimation

Genomic windows are correlated due to linked inheritance and shared genealogical history. We used block jackknife resampling[20] over non-overlapping 1 Mb blocks — a scale that

exceeds the typical LD decay distance of ∼500 kb in 1kGP populations[21] and provides approximately independent blocks for jackknife purposes. The genome was divided into 2,868 such blocks (defined as consecutive 1 Mb segments, irrespective of chromosome boundaries); blocks smaller than 500 kb at chromosome ends were excluded. Leave-one-block-out pseudovalues were used to compute jackknife standard errors and to construct confidence intervals for all group-level mean TMRCA estimates and their differences.

Demographic model simulations

To validate PCHMM calibration and to quantify the AFR−EAS mean TMRCA difference predicted under competing demographic hypotheses, we implemented three models in `msprime` v1.3[22]:

Model OoA (calibration reference). A standard three-population Out-of-Africa model[5]: ancestral population ($N_e = 7{,}300$; duration ∼200 ka), divergence of the proto-African lineage ($N_e = 12{,}300$), OoA bottleneck ($N_e = 2{,}100$; onset ∼65 ka), European–Asian split (∼21 ka), with growth to present-day sizes $N_{AFR} = 14{,}474$, $N_{EUR} = 34{,}039$, $N_{ASN} = 45{,}877$. Under this model, all modern humans share the same ancestral genealogy; within-population TMRCA depths differ only due to population size history, not ancestral structure.

Model A (panmictic + FitCoal bottleneck). A single panmictic ancestral population with a severe bottleneck parameterized following **(author?)**[7]: $N_e = 1{,}280$ for 117 ka beginning at ∼930 ka, flanked by $N_e = 100{,}000$ (before) and $N_e = 50{,}000$ (after), followed by a standard OoA split at 65 ka. Under this panmictic model, all modern populations are random draws from the same genealogy, and within-group TMRCAs must be equal regardless of population label.

Model D (structured ancestral population; Cobraa-like). Two ancestral populations that diverged at 1.5 Ma (proto-Africa lineage: $N_e = 15{,}000$; proto-OoA lineage: $N_e =$

10,000), with 20% gene flow from the proto-Africa lineage into the proto-OoA lineage at 300 ka (as inferred by **(author?)**[11]), followed by OoA exit at 65 ka with a bottleneck (N_e = 2,100). Under this model, modern Africans retain genealogical lineages from both ancestral populations, while modern non-Africans trace predominantly through the proto-OoA lineage, producing deeper within-AFR TMRCA.

Each model was simulated for 50 Mb using a uniform recombination rate of 10^{-8} per bp per generation, with three independent replicates per model (random seeds 42, 123, 456). Exact window TMRCA values were extracted from `msprime` tree sequences using the divergence matrix method. PCHMM estimates were computed from the resulting genotype matrices using the identical pipeline applied to real data. Calibration error was defined as $(\hat{T}_{\text{PCHMM}} - T_{\text{exact}})/T_{\text{exact}}$ for each group and each replicate.

Per-window TMRCA mixture analysis

To characterize the shape of the within-population TMRCA distribution — and specifically to test whether a fraction of East Asian genomic windows shows the shallow coalescence (<100 ka) that would be expected under a severe OoA bottleneck — we fitted mixture models to per-window $\hat{T}_G^{(w)}$ values aggregated across all 22 autosomes. We used Gamma mixture distributions (two-component Gamma mixture is flexible enough to capture both symmetric and skewed TMRCA distributions; exponential TMRCA under a constant-N_e population is a special case of Gamma). Expectation-maximization (EM) with 50 random restarts was used for parameter estimation; the optimal number of components $K \in \{1, 2, 3\}$ was selected by Bayesian Information Criterion (BIC), evaluated on every-other-block subsample to reduce spatial autocorrelation.

Under Model A, 31.4% of within-EAS windows would be expected to show coalescence at ≈ 55 ka (corresponding to the bottleneck epoch) — a prediction we use as a quantitative benchmark against the observed minimum mixture component location in the within-EAS distribution.

Background selection robustness

To evaluate whether background selection (BGS) could explain the AFR>EAS TMRCA pattern, we identified putatively neutral windows located >50 kb from any annotated gene transcript (UCSC hg38 RefGene track, downloaded August 2020 **(author?)**[23]). A 50 kb flanking buffer is standard for BGS exclusion[24] and encompasses the spatial scale at which BGS effects on diversity are well documented in humans. After merging overlapping buffered regions, 30.7% of the autosomal genome was retained as neutral (range: 8.0% for chr19 to 42.6% for chr22); this fraction is consistent with prior BGS masking analyses. The AFR−EAS mean TMRCA difference was recomputed using block jackknife restricted to neutral windows only, and compared to the genome-wide estimate.

Mutation rate sensitivity and AMR stratification

To assess sensitivity to the assumed mutation rate, we rescaled all TMRCA estimates by μ_{ref}/μ for $\mu \in \{1.0, 1.2, 1.4, 1.6\} \times 10^{-8}$ per bp per generation, spanning the range reported in recent large-scale de novo mutation studies[25,18]. Because TMRCA estimates scale as $\hat{T} \propto 1/\mu$, the within-group ratio $\hat{T}_{\text{AFR}}/\hat{T}_{\text{EAS}}$ is analytically invariant to the choice of μ; we verify this numerically. For AMR stratification, we recomputed mean TMRCA separately for each fine-grained AMR population (ASW, ACB, MXL, PUR, CLM, PEL) using the same group-level aggregate estimator, to examine how modern admixture composition is reflected in genealogical depth.

Computational implementation

Likelihood computations were implemented in JAX[26] with JIT compilation and batch vectorization across haplotype pairs. Genome-wide analysis of all 22 autosomes (240 samples; $\sim 2.9 \times 10^4$ unique haplotype pairs per chromosome) was completed on a compute node with eight NVIDIA L20 GPUs (48 GB VRAM each) and 128 CPU threads, requiring approxi-

mately 48 computing hours. All analysis code is available at [CITATION NEEDED].

Data availability

1000 Genomes Project Phase 3 high-coverage data are publicly available at `ftp.1000genomes.ebi.ac.uk/v`
Sample ancestry annotations are provided by the 1kGP consortium at the same location.

Acknowledgements

We thank the 1000 Genomes Project Consortium for making whole-genome data publicly available.

References

[1] Stringer, C. The origin and evolution of *Homo sapiens*. *Philosophical Transactions of the Royal Society B* **371**, 20150237 (2016).

[2] Hublin, J.-J. *et al.* New fossils from Jebel Irhoud, Morocco and the pan-African origin of *Homo sapiens*. *Nature* **546**, 289–292 (2017).

[3] Li, H. & Durbin, R. Inference of human population history from individual whole-genome sequences. *Nature* **475**, 493–496 (2011).

[4] Schiffels, S. & Durbin, R. Inferring human population size and separation history from multiple genome sequences. *Nature Genetics* **46**, 919–925 (2014).

[5] Gutenkunst, R. N., Hernandez, R. D., Williamson, S. H. & Bustamante, C. D. Inferring the joint demographic history of multiple populations from multidimensional SNP frequency data. *PLOS Genetics* **5**, e1000695 (2009).

[6] Terhorst, J., Kamm, J. A. & Song, Y. S. Robust and scalable inference of population history from hundreds of unphased whole genomes. *Nature Genetics* **49**, 303–309 (2017).

[7] Hu, W. *et al.* Genomic inference of a severe human bottleneck during the early to middle pleistocene transition. *Science* **381**, 979–984 (2023).

[8] Malaspinas, A.-S. *et al.* A genomic history of aboriginal australia. *Nature* **538**, 207–214 (2016).

[9] Mallick, S. *et al.* The simons genome diversity project: 300 genomes from 142 diverse populations. *Nature* **538**, 201–206 (2016).

[10] Cousins, T. & Durvasula, A. Insufficient evidence for a severe bottleneck in humans during the early to middle pleistocene transition. *Molecular Biology and Evolution* **42**, msaf041 (2025).

[11] Cousins, T., Scally, A. & Durbin, R. A structured coalescent model reveals deep ancestral structure shared by all modern humans. *Nature Genetics* (2025).

[12] Byrska-Bishop, M., Abel, H. J. *et al.* High-coverage whole-genome sequencing of the expanded 1000 genomes project cohort including 602 trios. *Cell* **185**, 3426–3440.e19 (2022).

[13] Prugnolle, F., Manica, A. & Balloux, F. Geography predicts neutral genetic diversity of human populations. *Current Biology* **15**, R159–R160 (2005).

[14] DeGiorgio, M., Jakobsson, M. & Rosenberg, N. A. Explaining worldwide patterns of human genetic variation using a coalescent-based serial founder model of migration outward from Africa. *Proceedings of the National Academy of Sciences* **106**, 16057–16062 (2009).

[15] Myers, S., Fefferman, C. & Patterson, N. Can one learn history from the allelic spectrum? *Theoretical Population Biology* **73**, 342–348 (2008).

[16] Tajima, F. Evolutionary relationship of DNA sequences in finite populations. *Genetics* **105**, 437–460 (1983).

[17] Watterson, G. A. On the number of segregating sites in genetical models without recombination. *Theoretical Population Biology* **7**, 256–276 (1975).

[18] Scally, A. The mutation rate in human evolution and demographic inference. *Current Opinion in Genetics & Development* **41**, 36–43 (2016).

[19] Moorjani, P. *et al.* A genetic method for dating ancient genomes provides a direct estimate of human generation interval in the last 45,000 years. *Proceedings of the National Academy of Sciences* **113**, 5652–5657 (2016).

[20] Künsch, H. R. The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* **17**, 1217–1241 (1989).

[21] Reich, D. E. *et al.* Linkage disequilibrium in the human genome. *Nature* **411**, 199–204 (2001).

[22] Kelleher, J., Etheridge, A. M. & McVean, G. Efficient coalescent simulation and genealogical analysis for large sample sizes. *PLOS Computational Biology* (2016).

[23] Navarro Gonzalez, J. *et al.* The UCSC genome browser database: 2021 update. *Nucleic Acids Research* **49**, D1046–D1057 (2021).

[24] McVicker, G., Gordon, D., Davis, C. & Green, P. Widespread genomic signatures of natural selection in hominid evolution. *PLOS Genetics* **5**, e1000471 (2009).

[25] Kong, A. *et al.* Rate of de novo mutations and the importance of father's age to disease risk. *Nature* **488**, 471–475 (2012).

[26] Bradbury, J. *et al.* JAX: composable transformations of Python+NumPy programs. `http://github.com/google/jax` (2018).